The weighted Dirichlet operator has infinitely many eigenvalues

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Task 1. Resolve Conjecture 6.1 of Monsalve-López and Seco [1]. For every $\alpha > 0$ and nonzero $a \in \mathbb{C}$, it asserts that the operator

$$V_a = M_{a-z}^* M_{a-z}$$

has empty point spectrum on the space with norm

$$\left\| \sum_{n\geq 0} f_n z^n \right\|_\omega^2 = \sum_{n\geq 0} |f_n|^2 \omega_n, \qquad \omega_n = \binom{n+\alpha}{n}.$$

The conjecture is false. A one-vector counterexample suffices, and a localized variational argument proves the opposite behavior throughout the stated parameter range.

Theorem 1. *For every $\alpha > 0$ and $a \neq 0$, the operator V_a has infinitely many isolated eigenvalues of finite multiplicity strictly above $(1+|a|)^2$, accumulating at $(1+|a|)^2$. Its essential spectrum is*

$$\sigma_{\mathrm{ess}}(V_a) = [(1-|a|)^2, (1+|a|)^2]. \tag{1}$$

In particular, for $\alpha = 1$ and $a = 1/4$ there is an eigenvalue $\lambda \geq 33/16 > 25/16$.

Weighted-shift representation. The monomials $e_n = z^n/\sqrt{\omega_n}$ form an orthonormal basis. Writing $S = M_z$, we have

$$Se_n = q_n e_{n+1}, \qquad q_n^2 = \frac{\omega_{n+1}}{\omega_n} = 1 + \frac{\alpha}{n+1}. \tag{2}$$

Hence S is bounded, $q_n > 1$, and $q_n \to 1$. Moreover,

$$V_a = |a|^2 I - \overline{a} S - a S^* + S^* S \tag{3}$$

is bounded, positive, and self-adjoint. If $Ue_n = e_{n+1}$, then $S - U$ is compact, since truncating its weight sequence gives finite-rank operators converging in operator norm. Likewise $S^*S - I$ is a compact diagonal operator. Thus V_a is a compact perturbation of

$$W_a = (|a|^2 + 1)I - \overline{a} U - a U^*.$$

The free half-line Jacobi operator $U + U^*$ has spectrum and essential spectrum $[-2, 2]$. A diagonal unitary removes the argument of a. Weyl invariance of the essential spectrum proves (1), also given by Theorem 1.2 of [1]. Every spectral point outside this interval is an isolated eigenvalue of finite multiplicity.

An exact counterexample. Since $e_0 = 1$ has norm one and $1, z$ are orthogonal,

$$\langle V_a e_0, e_0 \rangle = \|a - z\|_\omega^2 = |a|^2 + 1 + \alpha. \tag{4}$$

Its excess over the upper essential edge is $\alpha - 2|a|$. Whenever $\alpha > 2|a|$, the supremum of the spectrum therefore lies strictly above that edge and is an eigenvalue. For $\alpha = 1$, $a = 1/4$, the expectation is $33/16$ and the edge is $25/16$, with exact gap $1/2$. This already refutes the full universal conjecture.

Infinitely many eigenvalues for all parameters. Put $\rho = |a| > 0$ and $E = (1 + \rho)^2$. In a suitable diagonal unitary basis, the matrix of V_a has diagonal $1 + \rho^2 + \alpha/(n + 1)$ and positive off-diagonal entries ρq_n. For a real nonnegative finitely supported vector $u = (u_n)$ its quadratic form, denoted $Q(u)$, consequently satisfies

$$\begin{aligned} Q(u) - E\|u\|_2^2 &\geq \sum_{n\geq 0} \frac{\alpha u_n^2}{n+1} - 2\rho \sum_{n\geq 0} u_n^2 + 2\rho \sum_{n\geq 0} u_n u_{n+1} \\ &= \sum_{n\geq 0} \frac{\alpha u_n^2}{n+1} - \rho \sum_{n=-1}^{\infty} (u_{n+1} - u_n)^2, \qquad u_{-1} = 0. \end{aligned} \tag{5}$$

Here $q_n \geq 1$ and nonnegativity of the coordinates justify the first inequality; zero coordinates are used outside the support.

For an integer $L \geq 1$, place a tent vector on $[L, 3L - 2]$:

$$u_{L+j-1}^{(L)} = \min(j, 2L - j) \quad (1 \leq j \leq 2L - 1), \qquad u_n^{(L)} = 0 \text{ elsewhere.}$$

Direct summation gives

$$N_L := \|u^{(L)}\|_2^2 = L^2 + 2\sum_{j=1}^{L-1} j^2 = \frac{2L^3 + L}{3}, \qquad \sum_{n=-1}^{\infty} (u_{n+1}^{(L)} - u_n^{(L)})^2 = 2L. \tag{6}$$

On its support, $n + 1 \leq 3L$. Therefore (5) yields

$$\frac{Q(u^{(L)})}{N_L} - E \geq \frac{\alpha}{3L} - \frac{6\rho}{2L^2 + 1} > \frac{\alpha}{3L} - \frac{3\rho}{L^2} > 0 \quad \text{if } L > \frac{9\rho}{\alpha}. \tag{7}$$

Choose such an L_0 and put $L_j = 4^j L_0$. The supports of the corresponding tent vectors are separated by at least one zero coordinate. The vectors are orthogonal, and the Jacobi matrix has no quadratic-form cross terms between them. Thus, for every k, the span of the first k vectors is a k-dimensional subspace on which $Q(u) > E\|u\|_2^2$ for all nonzero u.

To see the spectral implication directly, if the spectral projection of V_a onto (E, ∞) had rank less than k, that subspace would contain a nonzero vector orthogonal to its range. The spectral theorem would give $Q(u) \leq E\|u\|_2^2$, a contradiction. The projection therefore has infinite rank. Since (E, ∞) is outside the essential spectrum, it consists of isolated eigenvalues of finite multiplicity. There must be infinitely many of them. Boundedness and the absence of accumulation outside the essential spectrum force their only accumulation point to be E. This proves Theorem 1. □

Scope and verification. The norm is exactly the one in the published conjecture. The paper's Theorem 1.5 treats a different, equivalent norm for $0 < \alpha < 1$; it does not assert this result for the displayed binomial weights. The general argument above is independent of finite matrix truncations. Exact checks verify the weight and coefficient normalization, the one-vector certificate, and the tent identities. Supplementary interval checks evaluate the actual Jacobi forms of selected tents.

Task 2. Determine the leading asymptotic decay of the eigenvalues above $(1+|a|)^2$, including its dependence on α and $|a|$, and obtain an effective error bound.

Task 3. Classify the eigenvalues below $(1-|a|)^2$ and determine how their existence and accumulation change as $|a|$ crosses one.

Task 4. Find sharp tail conditions on positive weights ω_n that ensure infinitely many eigenvalues above the free upper edge. Analyze the critical case $\omega_{n+1}/\omega_n - 1$ of order n^{-2}.

Task 5. Determine whether embedded or edge eigenvalues occur for the binomial weights, and describe the absolutely continuous and singular continuous parts of the spectral measure.

References

[1] M. Monsalve-López and D. Seco, *Towards spectral descriptions of cyclic functions*, Annals of Functional Analysis **16** (2025), article 52. Theorem 1.2, Section 2, and Conjecture 6.1. doi:10.1007/s43034-025-00446-0.